13.3.2 RJ45 Connector

The RJ45 connector is a small plastic cup which will be used to connect the wire inside the connector and ready to connect the Internet. The RJ45 connector looks similar like a telephone jack but it looks a slightly wider. The Ethernet cables are sometime called as RJ45 cables. In RJ45 the "RJ" stands for the Registered Jack and the "45" simply refers to the number of interface standard in the cable. See Figure 13.9



Figure 13.9 RJ45 Connector

Each RJ45 connector has eight pins and connected to each end of the Ethernet cable. since it has 8-position, 8-contact (8P8C) modular plug, It is also known as 8P8C connector. These plugs (connector) are then inserted into Ethernet port of the network card.

Wiring schemes and colour codes of the connector

The RJ45 connector has eight small jack inside to connect eight small wires of the patch cable. The eight cables are in eight different colors. Let's discuss that eight colors and where does that eight colors connect to the RJ45 connector.

Wiring schemes specifies how the wires to be connected with RJ45 connector. There are two wiring schemes available to terminate the twisted-pair cable on each end, which are T-568A and T-568B. The Table 13.1 describes the pinouts and colour coding of these schemes.

Table 13.1 Wiring schemes with colour codes

Pin	T-568A	Pin	T-568B
1	white / green stripe (Tx+)	1	white / orange stripe (Tx+)
2	green (Tx-)	2	orange (Tx-)
3	white / orange stripe (Rx+)	3	White / green stripe (Rx+)
4	blue	4	blue
5	white / blue stripe	5	white / blue stripce
6	orange (Rx-)	6	green (Rx-)
7	white / brown stripe	7	white / brown stripe
8	brown	8	brown

Although four pairs of wires are available in the cable, Ethernet uses only two pairs: Orange and Green. The other two colors (blue and brown) can be used ISDN or phone connections.

As we have already discussed about the pin details in the Ethernet connector. Now let's see the pin details of the Ethernet ports, the pin 1 shows the transmission positive the pin 2 shows the transmission negative the pin 3 shows the receiver positive. The pin 4 shows the reserved position it is nothing but tells the connection for some other use. The pin 5, 7, 8 are also used in reserved position. The pin 6 shows the receiver negative. The two main signals of the pins: the one is the TX which is transmission of data and RX

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